Identification of an Unknown - “A”

Georgia Blaze

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Chem 2081- 001

Introduction

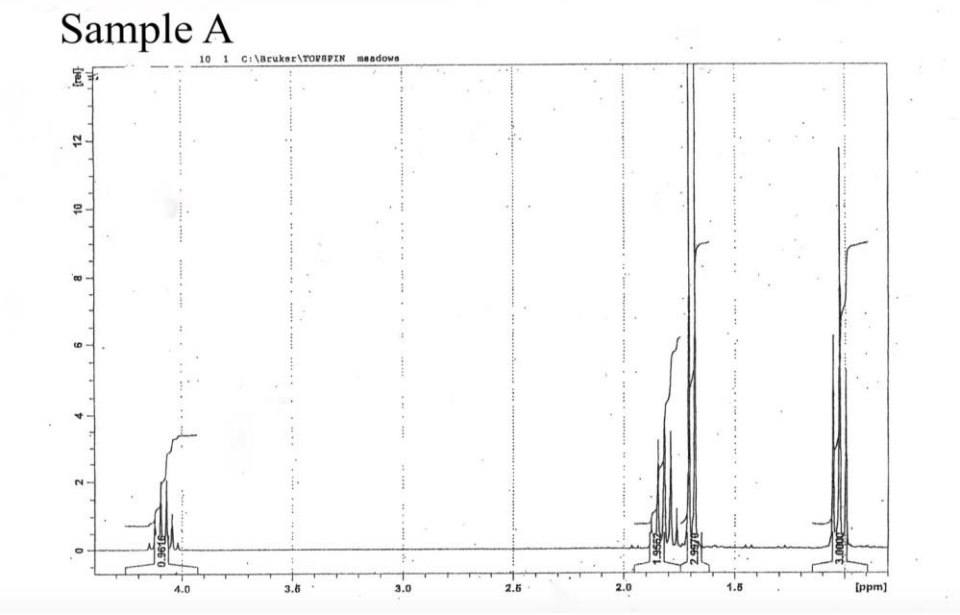
In this formal lab multiple different factors are being used to identify an unknown. The goal of this experiment is to identify an unknown person using three different types of spectral data sheets. The three sheets that were given are NMR, FTIR, and MS. Motivation was used to use these sheets and find the unknown. This is a necessary skill because motivation is the reason or reasons one has for acting or behaving in a particular way.

Materials and Methods

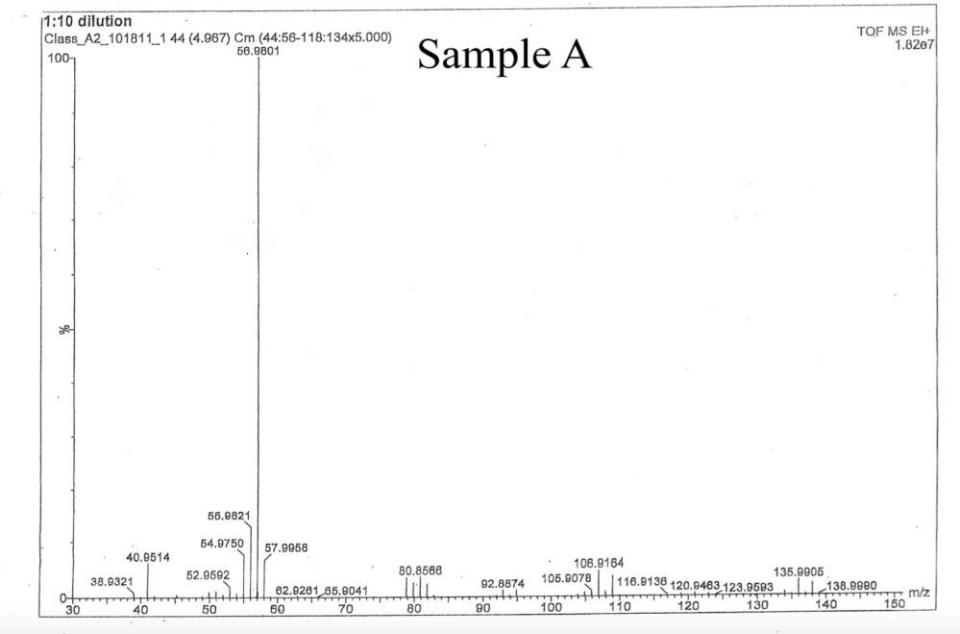
NMS instruments are continuous wave instruments that work by irritating the sample with radio frequency energy. When there is a match between the RF energy and the energy difference between the two spins states of the nucleus a signal appears and is detected. The high magnetic fields necessary for these instruments are accomplished by using superconducting electromagnets. These magnets can only be used at low temperatures by keeping the magnets in double-jacketed Dewar vessels cooled by liquid helium and liquid nitrogen. The next method is FTIR. FTIR is infrared (IR) spectroscopy which is one of the important techniques for finding the compound structure of organic molecules. This type of method is an effective way to find presence of absence of functional groups. In this method infrared energy passes through a sample of organic compound. In this IR spectrum, energy is measured as wavelengths that are plotted horizontally and the intensity of the energy is plotted vertically. The different bands in these graphs can show charcteristics of certain functional groups that will help identify the organic molecule. The last method that was used is the mass (MS) spectrometers method. The MS method is when a compound is introduced into the mass spectrometer and is changed into a gas

phase ion through ionization technique. The ions are then sorted by their mass-to-charge ratios by a mass analyzer and then the ions generate an electric current which leads to the mass spectrum. This then corresponds to the molecular weight of the organic compound that can help identify it.

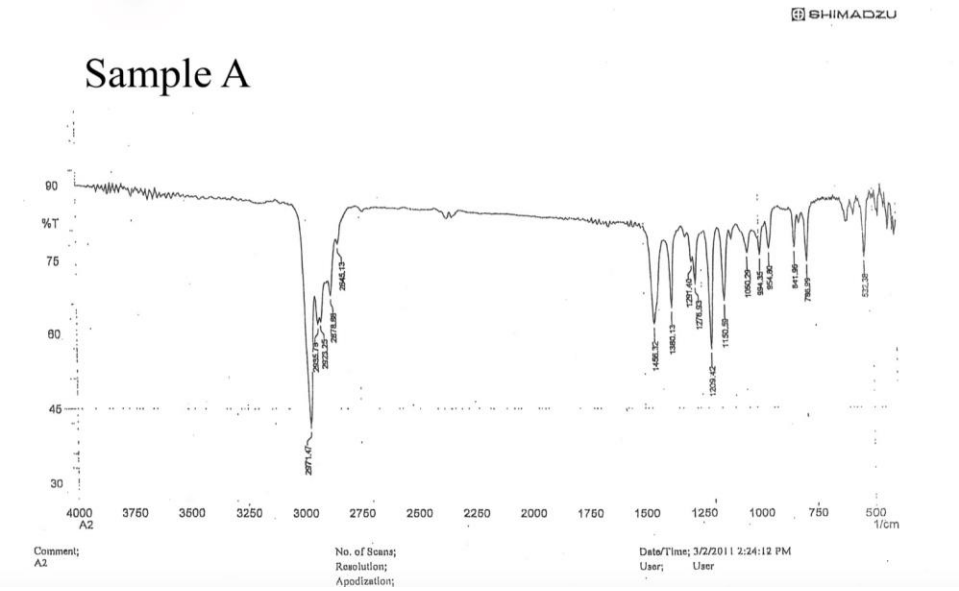
NMR Sample A



GCMS Sample A

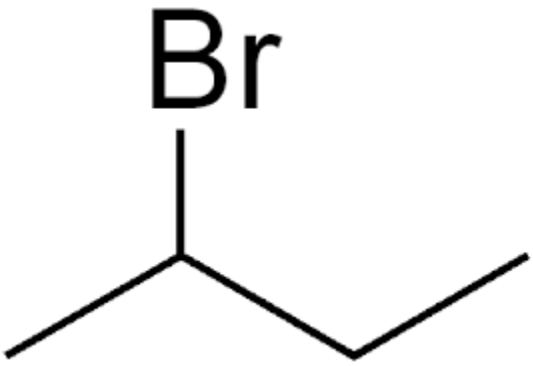


FTIR Sample A



Conclusions

When using the NMR method for this unknown compound “A”, there is quintet of the integration at 1.9552, a doublet at 2.9970, a triplet at 3.000, and a sextet at 0.9616. These numbers give peaks on the graph that show the number of hydrogens neighboring the bromine and also the distance of the peaks relative to the distance from the bromine. With what is given in these spectra and identifying these peaks makes the conclusion of this organic molecule 2-bromobutane.



References

2022 CHEM 2081 Organic Chemistry 2 Lab Manual. Auburn, (AL): Auburn University